

Small Is Beautiful: Geometric Composition for CPU-Only Document Conversion at 251 MB

Supplementary Material

Anonymous WACV Applications Track submission

Paper ID *****

Table S1. By document type (OmniDocBench v1.6 full, 1651 pages). Text and Order are normalized edit distances (lower better); CDM is Character Detection Matching on display formulas and TEDS is Tree-Edit-Distance Similarity on tables (higher better). A dash marks a stratum with no formula or table content.

Type	N	Text↓	CDM↑	TEDS↑	Order↓
research_report	132	0.014	–	84.4	0.285
magazine	149	0.056	–	77.0	0.103
PPT2PDF	253	0.066	88.6	86.1	0.105
academic_literature	215	0.080	89.1	75.0	0.127
book	276	0.083	83.7	88.9	0.163
colorful_textbook	159	0.097	82.0	87.4	0.155
note	118	0.101	75.9	60.0	0.103
newspaper	151	0.103	–	71.1	0.155
exam_paper	193	0.127	91.2	91.5	0.148
historical_document	5	0.560	–	–	0.432

Table S2. By language (OmniDocBench v1.6 full).

Language	N	Text↓	CDM↑	TEDS↑	Order↓
english	755	0.070	90.9	77.9	0.126
en_ch_mixed	116	0.080	93.3	84.6	0.143
simplified_chinese	765	0.098	77.1	84.4	0.184
traditional_chinese	13	0.213	–	73.9	0.324
other	2	0.084	–	–	0.118

Table S3. By layout (OmniDocBench v1.6 full).

Layout	N	Text↓	CDM↑	TEDS↑	Order↓
1-and-more column	155	0.063	93.8	88.8	0.117
double column	184	0.069	90.1	87.4	0.107
single column	887	0.074	86.8	80.8	0.154
three column	53	0.098	84.9	69.7	0.164
other layout	372	0.121	74.3	80.0	0.161

001 This supplementary material provides (i) the full per-section
002 breakdown of PRISM’s OmniDocBench v1.6 run, (ii) the
003 complete Fox benchmark comparison, (iii) the full pertur-
004 bation audit summarized in the main paper, and (iv) repro-
005 ducibility details. All numbers are the same submitted build
006 reported in the main paper; no result here uses a different
007 model, additional tuning, or a newer benchmark version.

008 S1. Per-section OmniDocBench v1.6 break- 009 down

010 Tables S1 to S3 give PRISM’s full OmniDocBench v1.6
011 result (1651 pages, official harness) disaggregated by doc-
012 ument type, language, and layout. CDM here is a page-
013 average; the headline in the main paper is the harness’s
014 pooled per-formula mean, which weights formula-dense
015 pages more heavily. These are the disaggregated compo-
016 nents of the 87.29 composite reported in the main paper.

S2. Full Fox benchmark comparison

Table S4 is the complete Fox benchmark comparison; the main paper reports the headline figures only. Reference rows are as published in the Fox paper; all comparison systems are GPU generative models evaluated natively. PRISM is run zero-shot and scored on plain text after stripping its structural markup, so the comparison understates it: the baselines emit plain text natively while PRISM discards structure it recovered correctly.

S3. Perturbation audit: how much of the score is emission convention?

We applied semantically-null transformations to our final predictions, from which a human reader would extract identical content, and re-scored with the untouched official harness (Tab. S5). The harness proves *robust* where content stays faithful: markdown styling is normalized away exactly, and even re-adding running headers/footers ver-

Table S4. Fox benchmark page OCR, full comparison. EN and ZH are normalized edit distance against the reference transcription (lower is better) on the English and Chinese splits. PRISM is scored after stripping structural markup the baselines never produce.

Model	Size	EN↓	ZH↓
InternVL-ChatV1.5 [3]	26B (GPU)	0.393	0.265
Vary [6]	7B (GPU)	0.092	0.113
DocOwl1.5 [4]	7B (GPU)	0.258	—
Fox [5]	1.8B (GPU)	0.046	0.061
Vary-toy [7]	1.8B (GPU)	0.082	0.142
GOT-OCR2.0 [8]	580M (GPU)	0.035	0.038
Nougat [2]	250M (GPU)	0.255	—
Qwen-VL-Max [1]	API	0.057	0.091
Qwen-VL-Plus [1]	API	0.096	0.121
PRISM (ours)	251 MB (CPU)	0.075	0.093

Table S5. Emission-convention perturbation study: semantically-null transformations of our final predictions, re-scored with the unmodified official harness (1651 pages). Δ in points ($\times 100$).

Perturbation (content unchanged)	Δ Text	Δ Order
bold-wrap first sentence of every 3rd para	0.00	0.00
re-add GT marginalia verbatim	+0.06	-0.01
split paragraphs at sentence boundaries	+0.41	+1.54

batim costs only 0.06 text points, because the matcher pairs them with the annotated-but-unscored ground-truth elements. The sensitivity is elsewhere: **segmentation granularity**. Emitting the same text as sentence-level blocks instead of paragraphs costs 0.41 text points and 1.54 reading-order points (0.14 composite points) purely by convention. Combined with the interaction bugs we repaired during development (a line-break rewrite corrupting matrix row separators in 28% of matched formula pairs; *noisy OCR*’d marginalia, unlike verbatim ones, costing 0.36 text points through failed pairing), the picture is consistent: benchmark scores reflect both parsing ability and emission or segmentation conventions at the tenth-of-a-point composite scale and the full-point component scale, comparable to many published leaderboard gaps. We recommend perturbation audits of this kind as standard practice when comparing document parsers.

S4. Reproducibility

All PRISM numbers are produced by the public OmniDocBench harness at its default configuration (quick-match matcher; CDM compiled with the harness’s pinned TeX Live 2025 image). Accuracy is always reported CPU-only; the GPU measurements in the main paper affect latency only and never the emitted text.

Model inventory. Six ONNX inference graphs, 251 MB total (decimal MB throughout): layout detector (PP-DocLayoutV3, RT-DETR, 130.5 MB), text OCR stack (PP-OCRv6, 31.1 MB), Texo formula recognizer (81.9 MB, 20.4M parameters, fp32), and SLANet-plus table model (7.4 MB). No component exceeds the layout detector; no PyTorch or Paddle runtime is resident. Per-graph checksums are in the code release.

Measurement configuration. All timing on an Intel Core i7 consumer machine (16 logical cores), Windows 11, onnxruntime CPU execution provider, no quantization applied to any graph (all six are fp32). Full-benchmark latency uses the dual-worker configuration; the thread-scaling curve and the 100-page head-to-head use single-worker, so their per-page numbers are not directly comparable to the dual-worker headline. Peak RAM is the maximum working set of the whole process tree including all workers.

Benchmark harnesses. OmniDocBench v1.6 (1651 pages) and v1.5 are each scored with their own unmodified official harness against their own release annotations. olmOCR-Bench uses commit 54a96a6, zero-shot, with PDFs rasterized at 200 DPI, over all assertions including the per-document baseline category. Fox uses the 212-page EN+CN split with structural markup stripped before edit-distance scoring.

Thresholds. Detection gates are 0.30 (formula) / 0.50 (all other classes); the empty-page rescue triggers below 30 content characters; the probe accepts a photometric correction only at a $\geq 2\%$ confident-text-surface improvement. A formula-gate sensitivity sweep (main paper) confirms the composite is flat across $[0.20, 0.40]$.

Normalization internals. Three open-loop details remain load-bearing inside the accepted corrections. Glare inpainting precedes shadow flattening (flattening whitens the page, which a global-brightness glare detector would then inpaint wholesale) and is skipped when the mask exceeds 20% of the page, since true specular glare is local. A rectification quadrilateral is warped only if convex, with corner angles in $[50^\circ, 130^\circ]$, and *containing the document* ($\geq 55\%$ of bright-region mass inside, interior $\geq 35\%$ bright); without the containment gate, wood-grain backgrounds produce confident quads that destroy the page. Frequency-domain moiré removal is enabled only for photographed screens: on paper, the dominant high-frequency spikes *are* the text-line periodicity. Recognizer crops are always taken from a pre-destructive copy, so OCR sees original pixels even when normalization informed detection.

Table S6. Complete model inventory (ONNX inference graphs), decimal MB. Total 250.9MB across six graphs. No component exceeds the 130.5 MB layout detector; the two generative recognizers (Texo) total 20.4M parameters, fp32.

Stage	Model	Size
Layout	PP-DocLayoutV3 (RT-DETR)	130.5 MB
OCR detection	PP-OCRv6-small det [†]	9.9 MB
OCR recognition	PP-OCRv6-small rec	21.2 MB
Formula encoder	Texo (distilled)	54.2 MB
Formula decoder	Texo (distilled)	27.7 MB
Table structure	SLANet-plus	7.4 MB
Total		250.9 MB

[†] Also serves as the verification probe.

Table S7. Full-run efficiency (1651 pages, 16-logical-core CPU, dual-worker benchmark configuration). Latency is end-to-end per page including all specialist workers; RAM is the summed resident set of the whole process tree, sampled at 0.3 s.

Latency (s/pg)		RAM (GB)	
median	5.91	median	1.04
mean	7.16	p90	1.13
p90	12.71	p95	1.21
p95	15.56	p99	1.34
p99	19.99	peak	1.70

Code. The anonymized supplementary archive submitted with this paper contains the complete pipeline source, every configuration file, and the experiment log, together with environment setup and run instructions reproducing every number in this paper. It is the exact build reported throughout; no result here uses a later or retuned version.

References

- [1] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-VL: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023. 2
- [2] Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. Nougat: Neural optical understanding for academic documents. *arXiv preprint arXiv:2308.13418*, 2023. 2
- [3] Zhe Chen, Weiyun Wang, Hao Tian, Shenglong Ye, Zhangwei Gao, Erfei Cui, Wenwen Tong, Kongzhi Hu, Jiapeng Luo, Zheng Ma, et al. How far are we to GPT-4V? closing the gap to commercial multimodal models with open-source suites. *arXiv preprint arXiv:2404.16821*, 2024. 2
- [4] Anwen Hu, Haiyang Xu, Jiabo Ye, Ming Yan, Liang Zhang, Bo Zhang, Chen Li, Ji Zhang, Qin Jin, Fei Huang, et al. mPLUG-DocOwl 1.5: Unified structure learning for ocr-free document understanding. *arXiv preprint arXiv:2403.12895*, 2024. 2

Table S8. Synthetic-defect study: mean per-block OCR edit distance (lower is better) on 40 text-heavy benchmark pages (20 EN / 20 ZH) under harsh photometric degradations with valid ground truth, with 95% bootstrap confidence intervals over pages. *none* = no correction, *open* = open-loop camera stack, *verified* = probe-gated stack. The verified arm tracks the no-correction arm to within confidence in every condition except glare, recovering the near-total rejection policy automatically, while the open-loop arm pays a tax in four of five. The gate's function is to make a correction stack safe to ship where one is required for geometric rectification, not to improve on doing nothing photometrically: that equivalence is the result.

Input	none	open-loop	verified
clean	0.177	0.184	0.185
shadow (0.18×)	0.181	0.185	0.188
glare	0.187	0.197	0.199
tungsten cast	0.187	0.183	0.185
shadow + cast	0.180	0.185	0.187

95% paired-bootstrap CI on (verified — none), all conditions: straddles zero except glare +0.011 [0.000, 0.027].

Table S9. Real captures: 44 uncontrolled defect photographs (hand shadows, tungsten lamps, page curl, deep corner shadows), three arms, paired bootstrap over pages. *none* = corrections disabled, *open* = open-loop stack, *verified* = probe-gated stack. Open-loop is significantly worse than no correction; the gate recovers that loss and is indistinguishable from leaving the page untouched, bounding the cost of shipping the correction stack at roughly one point.

Contrast	Δ yield ratio [95% CI]
open — none	−0.0950 [−0.1660, −0.0403] *
verified — none	−0.0037 [−0.0106, +0.0020]
verified — open	+0.0912 [+0.0371, +0.1601] *

*CI excludes zero.

- [5] Chenglong Liu, Haoran Wei, Jinyue Chen, Lingyu Kong, Zheng Ge, Zining Zhu, Liang Zhao, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Fox: Focus anywhere for fine-grained multi-page document understanding. *arXiv preprint arXiv:2405.14295*, 2024. 2
- [6] Haoran Wei, Lingyu Kong, Jinyue Chen, Liang Zhao, Zheng Ge, Jinrong Yang, Jianjian Sun, Chunrui Han, and Xiangyu Zhang. Vary: Scaling up the vision vocabulary for large vision-language models. In *ECCV*, 2024. 2
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(a) PP-DocLayoutV3 detection

have n dis-
at least 2,
of general-
ing $P(x)$, so
omputation

$R''(x)$.

t the great-
but $P(x) =$
ce (since O

and the other of which sums to $xy / ((1-y)(1-x))$
Therefore

$$\begin{aligned} \text{Formula } S(x,y) &= \frac{\text{conf } 0.93}{(1-x)(1-y)} - \frac{x^3y}{(1-x)(1-x^2y)} \\ &= \frac{xy^3}{(1-y)(1-xy^2)} \\ &= \frac{xy(1+x+y+xy-x^2y^2)}{(1-x^2y)(1-xy^2)} \end{aligned}$$

and the desired limit is

(b) Otsu ink projection, row-band splitting

split: gap > max(6 px, 0.25 * row h) gap NOT split: vertical ink bridge

row ink mass

$$S(x,y) = \frac{xy}{(1-x)(1-y)} - \frac{x^3y}{(1-x)(1-x^2y)}$$

$$= \frac{xy(1+x+y+xy-x^2y^2)}{(1-x^2y)(1-xy^2)}$$

(d) Texo decode, compiled

with geometry layer — per-line CDM 0.90 / 0.77 / 0.97

$$\begin{aligned} S(x,y) &= \frac{xy}{(1-x)(1-y)} - \frac{(1-x)(1-x^2y)}{(1-y)(1-xy^2)} \\ &= \frac{xy(1+x+y+xy-x^2y^2)}{(1-x^2y)(1-xy^2)} \end{aligned}$$

naive whole-box decode (geometry layer off)

$$\begin{aligned} &\langle 1 - \chi \downarrow 1 - \chi V^- \rangle \\ &xV(1+x+v+xy-x^2v^2) \\ &1 - x^2v \ln 1 - xv^2 \end{aligned}$$

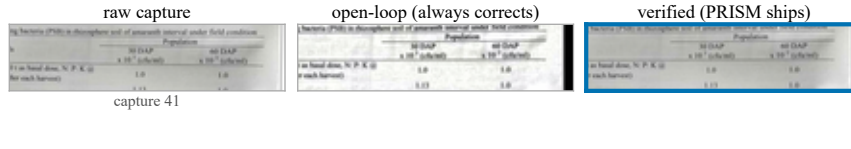
(c) band chunks: fraction fusion + trimming

B0: $\frac{xy}{(1-x)(1-y)}$ B1: $\frac{x^3y}{(1-x)(1-x^2y)}$ isolated chunks in B0
trimmed (speck + detached numerator)

B2: $\frac{xy^3}{(1-y)(1-xy^2)}$

B3: $= \frac{xy(1+x+y+xy-x^2y^2)}{(1-x^2y)(1-xy^2)}$ num / bar / den
re-fused into B3

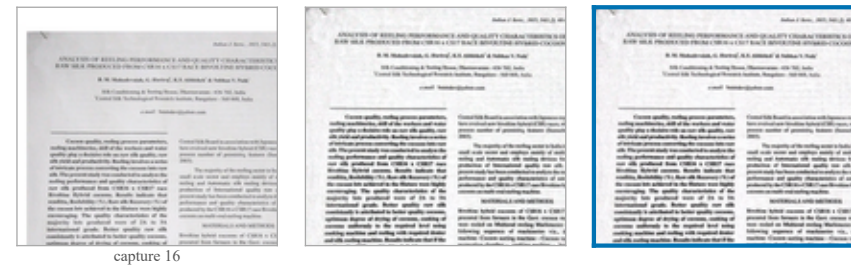
Figure S2. The deterministic formula-geometry layer on a real OmniDocBench detection. (a) PP-DocLayoutV3 emits one Formula box spanning a multi-line derivation. (b) Ink is binarized (Otsu) and projected onto rows; bands split at gaps exceeding $\max(6 \text{ px}, 0.25 \times \text{row height})$, except where a vertical ink bridge spans the gap. (c) The fraction triple is re-fused into a single band and isolated chunks are trimmed. (d) Texo decodes each band (per-line CDM 0.90 / 0.77 / 0.97); decoding the whole box naively, without the geometry layer, produces the malformed output below. The layer adds zero parameters.

Recognition-verified normalization on uncontrolled captures — probe: DBNet confident-text surface, accept if $\times \geq 1.02$ (fixed build)

ACCEPTED

rectify $\times 1.03$

rectify verified; probe rejects the shadow and clahe

raw: $\times 10^3$ (cfu/ml)open: $\times 10^3$ (cfu/ml)verif: $\times 10^3$ (cfu/ml)

ACCEPTED

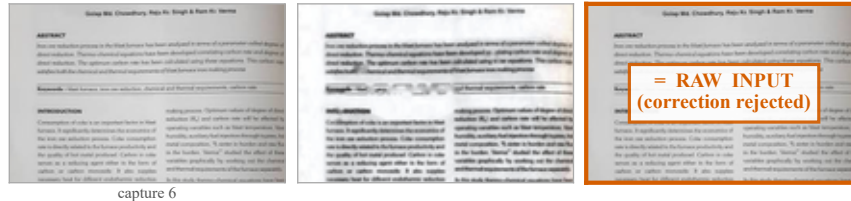
rectify $\times 1.13 \cdot \text{shadow} \times 1.03$

pixel routing called it clean; probe rescues the correction

raw: Bivoltine Hybrid cocoons. Results

open: Bivoltine Hybrid cocoons. Results

verif: Bivoltine Hybrid cocoons. Results



REJECTED

shadow flatten, probe $\times 0.13$

shadow-divide erases strokes; gate ships the raw input

raw: and thermal requirements of tse furace

open: and the nal sog an m e yes

verif: and thermal requirements of tse furace



REJECTED

shadow stats $0.98 \cdot \text{probe} \times 1.01$

slide gradient mimics a shadow; probe leaves it alone

raw: the west nor for the MNČs of the

open: the west nor for the MNČs of the

verif: the west nor for the MNČs of the

Figure S3. The recognition-verification gate on uncontrolled camera captures. For accepted proposals the gate ships the corrected crop; for rejected proposals it ships the raw input unchanged. Row 3 is the case an open-loop stack cannot avoid: the shadow divide's background estimate contains the text ink, erasing strokes, and the probe rejects it at $0.13 \times$. Row 4 is a presentation slide whose gradient scores higher on shadow statistics than genuine shadows do; the probe leaves it alone.

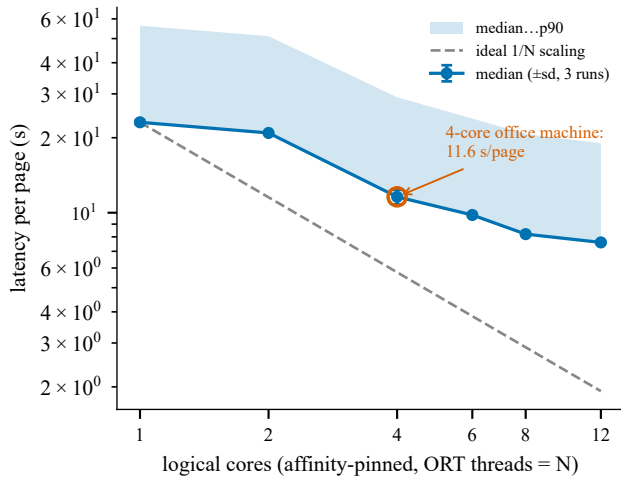


Figure S4. Median per-page latency vs. logical core count (affinity-pinned, ONNX-Runtime threads = N , single-worker, fixed stratified subset, 3 runs; accuracy identical at every point). Shaded band is median to p90. Most of the speedup arrives by 4 cores (11.6 s/page, circled); the curve then departs the ideal $1/N$ line (dashed) as the autoregressive formula decoder dominates. PRISM is the only system in the comparison for which this curve can be reported, being the only CPU-only one. Single-worker measurement; the 5.9 s/page headline (Tab. S7) uses the dual-worker benchmark configuration.